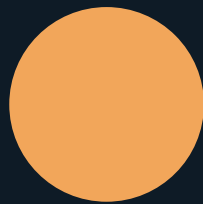


ROCKY MOUNTAIN RUBY · LIGHTNING TALK

# Async Workflows Durable Execution

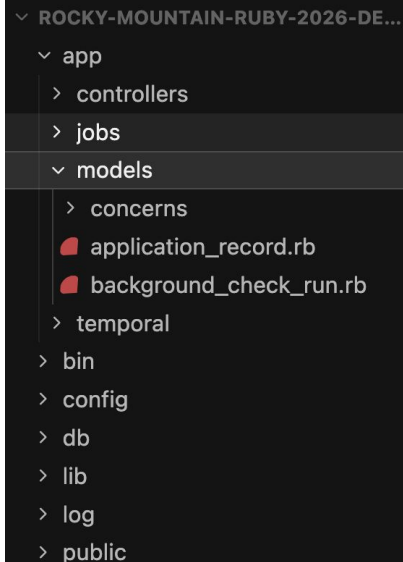
---



# Election theme

| Candidate            | Async Job Library                                                                                                                                                          | Open Source License       |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>Incumbent</b>     | Sidekiq                                                                                                                                                                    | MIT                       |
| <b>Challenger</b>    | <b>Temporal</b>                                                                                                                                                            | MIT                       |
| <b>Up and comers</b> | Postgres or SQLite-backed Durable Execution <ul style="list-style-type: none"><li>- DBOS, pg_durable by Microsoft</li><li>- ResonateHQ, Absurd Workflows, Obelis</li></ul> | MIT / Apache 2.0 / Varies |

# Demo Project: Background Check



```
▼ ROCKY-MOUNTAIN-RUBY-2026-DE...  
  ▼ app  
    > controllers  
    > jobs  
    ▼ models  
      > concerns  
      application_record.rb  
      background_check_run.rb  
    > temporal  
  > bin  
  > config  
  > db  
  > lib  
  > log  
  > public
```

- 1 Find Former Names
- 2 Find Jurisdictions
- 3 Send Searches to Counties
- 4 Evaluate Results
- 5 Compare Customer Assessments
- 6 Request Candidate Story
- 7 Notify Customer And Candidate
- 8 Handle Possible FCRA Dispute

INCUMBENT

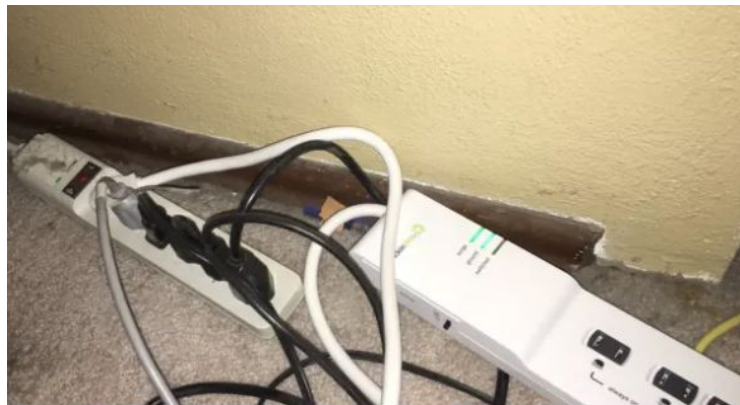
# Sidekiq

## Pros

- Durability / retries
- Well known- started in 2012
- Flat cost (Redis, Ruby workers, Sidekiq Pro)

## Cons

- Difficult to trace one Workflow
- ... across many Jobs



*"Daisy Chain"*

INCUMBENT

# Sidekiq

| <a href="#">←</a> <a href="#">→</a> <a href="#">↺</a> <a href="#">🔍</a> <a href="#">http://localhost:3000/sidekiq/queues/default</a>                                                                         |                                                 |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|---------------------------------------|
| <b>Sidekiq</b> <a href="#">Dashboard</a> <a href="#">Busy</a> <a href="#">Queues</a> <a href="#">Retries</a> <a href="#">Scheduled</a> <a href="#">Dead</a> <a href="#">Metrics</a> <a href="#">Profiles</a> |                                                 |                                       |
| Current jobs in default 30                                                                                                                                                                                   |                                                 |                                       |
| # ↓                                                                                                                                                                                                          | Job                                             | Arguments                             |
| 30                                                                                                                                                                                                           | BackgroundCheck::FindJurisdictionsJob           | 10                                    |
| 29                                                                                                                                                                                                           | BackgroundCheck::CompareCustomerAssessmentsJob  | 10                                    |
| 28                                                                                                                                                                                                           | BackgroundCheck::SendSearchJob                  | 14, "Jane Doe", "Harris County, TX"   |
| 27                                                                                                                                                                                                           | BackgroundCheck::RequestCandidateStoryJob       | 14                                    |
| 26                                                                                                                                                                                                           | BackgroundCheck::CompareCustomerAssessmentsJob  | 7                                     |
| 25                                                                                                                                                                                                           | BackgroundCheck::EvaluateResultsJob             | 11                                    |
| 24                                                                                                                                                                                                           | BackgroundCheck::EvaluateResultsJob             | 14                                    |
| 23                                                                                                                                                                                                           | BackgroundCheck::RequestCandidateStoryJob       | 5                                     |
| 22                                                                                                                                                                                                           | BackgroundCheck::FindFormerNamesJob             | 5                                     |
| 21                                                                                                                                                                                                           | BackgroundCheck::SendSearchJob                  | 9, "Jane A. Smith", "Cook County, IL" |
| 20                                                                                                                                                                                                           | BackgroundCheck::RequestCandidateStoryJob       | 12                                    |
| 19                                                                                                                                                                                                           | BackgroundCheck::FindFormerNamesJob             | 9                                     |
| 18                                                                                                                                                                                                           | BackgroundCheck::BackgroundCheckOrchestratorJob | 12                                    |
| 17                                                                                                                                                                                                           | BackgroundCheck::RequestCandidateStoryJob       | 9                                     |

## CHALLENGER

# Temporal

Workflow: One thread to tie all Activities together

### Timeline ⓘ

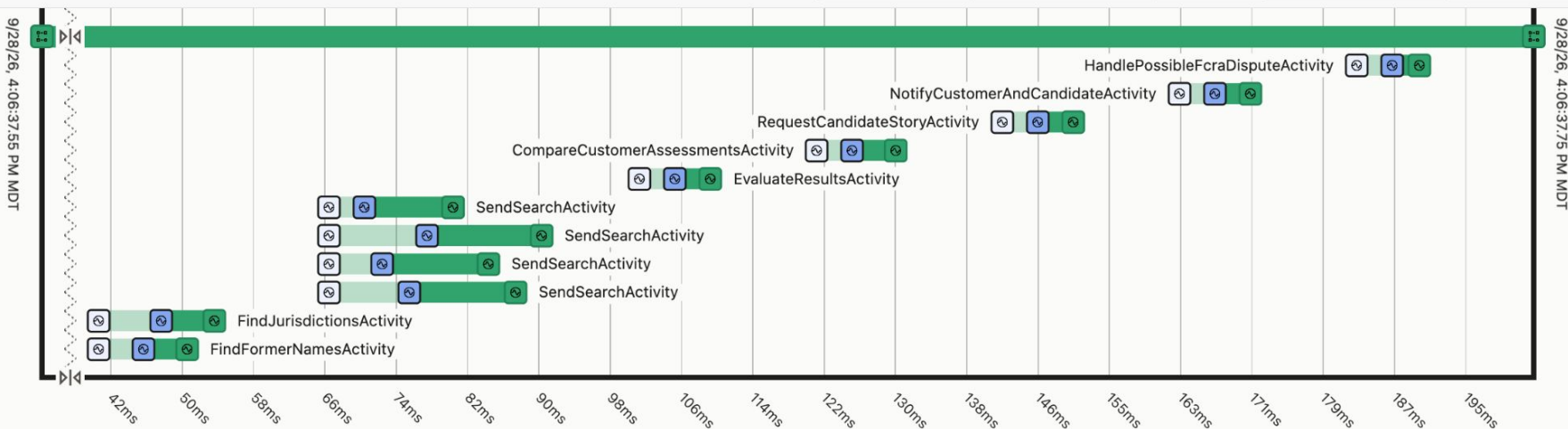
⌵ Descending

⏮ Show idle time

🔍 Filter

• Auto Refresh Off

📄 Download



## CHALLENGER

# Temporal

Clear Activities input and output

SendSearchActivity 18ms

Close X

15 Activity Task Scheduled

Sep 28, 2026, 4:06:37.61 PM MDT

Activity ID4

Activity TypeSendSearchActivity

Task Queue KindTASK\_QUEUE\_KIND\_NORMAL

Start To Close Timeout30 seconds

Workflow Task Completed Event ID13

Retry Policy Initial Interval1s

Retry Policy Backoff Coefficient2

Retry Policy Maximum Interval100s

Use Workflow Build IDtrue

Task Queue Namebackground-check

Input

"Jane A. Smith"

"Kings County, NY"

21 Activity Task Started

Sep 28, 2026, 4:06:37.62 PM MDT

Identity17430@PGFN2557NC

Request ID78053783-6f38-4966-94ff-f3074b4fc3ca

Attempt1

Worker Version53c2bd3db614829af4e0cb58a4d581bc

22 Activity Task Completed

Sep 28, 2026, 4:06:37.63 PM MDT

Identity17430@PGFN2557NC

Result"Search complete for Jane A. Smith in Kings County, NY: no records found"

## CHALLENGER

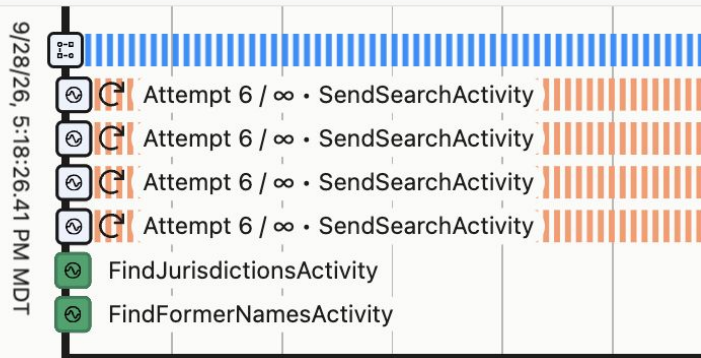
# Temporal

Durability: Configurable retries

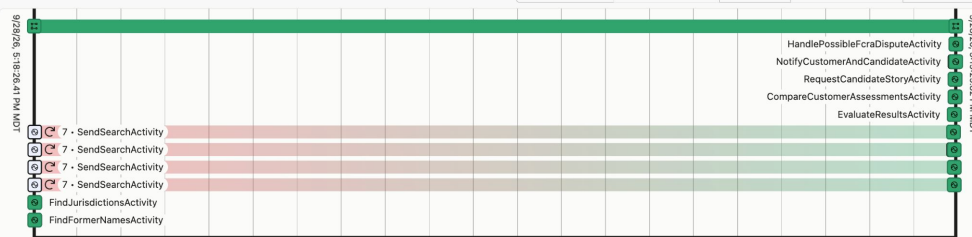
Failing Activity due to bug or data furnisher API

Activity succeeded on 7th retry

### Timeline ⓘ



### Timeline ⓘ



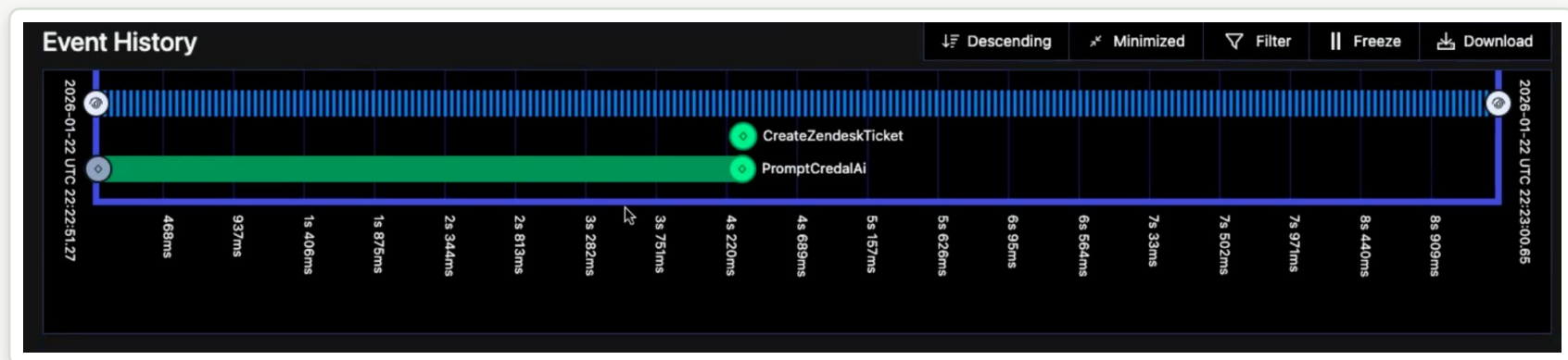


## CHALLENGER

# Temporal

Managing Agent Workflows

Failing API requests / remembering AI conversation context and ability to resume days or weeks later



## CHALLENGER

# Temporal

Cross Language Support (Rust worker 😊)

### 53 Activity Task Scheduled

|                                  |                                   |
|----------------------------------|-----------------------------------|
| Activity ID                      | 11                                |
| Activity Type                    | HandlePossibleFcraDisputeActivity |
| Task Queue Kind                  | TASK_QUEUE_KIND_NORMAL            |
| Start To Close Timeout           | 30 seconds                        |
| Workflow Task Completed Event ID | 52                                |
| Retry Policy Initial Interval    | 1s                                |
| Retry Policy Backoff Coefficient | 2                                 |
| Retry Policy Maximum Interval    | 100s                              |
| Task Queue Name                  | background-check-rust             |

Input

"Harry Kane"

```
#[activity(name = "HandlePossibleFcraDisputeActivity")]
pub(crate) async fn handle_possible_fcra_dispute(
    _ctx: ActivityContext,
    candidate_name: String,
) -> Result<String, ActivityError> {
    Ok(format!(
        "FCRA dispute check for {candidate_name}: no dispute filed, process complete (via Rust worker)"
    ))
}

#[tokio::main]
async fn main() -> Result<(), Box<dyn std::error::Error>> {
    let raw_address =
        std::env::var("TEMPORAL_ADDRESS").unwrap_or_else(|_| "localhost:7233".to_string());
    let address = if raw_address.starts_with("http") {
        raw_address
    } else {
        format!("http://{raw_address}")
    };
};
```

# Temporal w/ Traces to Honeycomb.io

Overview BubbleUp Correlations Traces Explore Data

Shows up to 10 traces with the slowest spans from the selected time range. [Learn more.](#) 

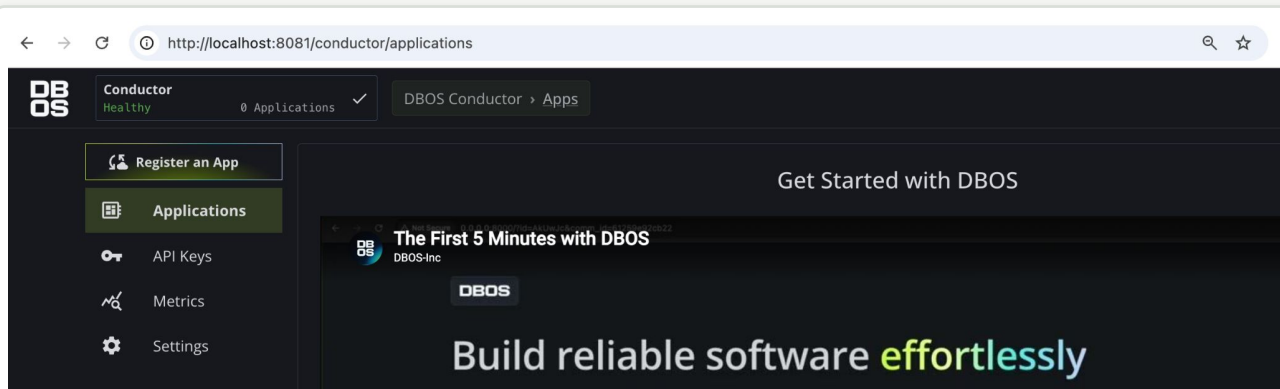
|                                                                                   | Root Service             | Root Name                             | Root Duration (ms) | Spans | Summary                                                                             | Trace ID                         |
|-----------------------------------------------------------------------------------|--------------------------|---------------------------------------|--------------------|-------|-------------------------------------------------------------------------------------|----------------------------------|
|  | rocky-mountain-ruby-demo | StartWorkflow:BackgroundCheckWorkflow | 4.613              | 12    |  | 2e278a181d4c6427c3cdac7aee2bb33f |
|  | rocky-mountain-ruby-demo | RunActivity:SendSearchActivity        | 0.01               | 1     |  | c8db69c526e5970f5282b91bb30bfbe5 |
|  | rocky-mountain-ruby-demo | RunActivity:SendSearchActivity        | 0.008              | 1     |  | 038e998effb69e888b0a31e8d5750afe |
|  | rocky-mountain-ruby-demo | RunActivity:SendSearchActivity        | 0.011              | 1     |  | f0db9f6ec6378c2ef14b028923b154b3 |
|  | rocky-mountain-ruby-demo | RunActivity:SendSearchActivity        | 0.014              | 1     |  | 1fec33ac1a956d75f98296563e785226 |
|  | rocky-mountain-ruby-demo | RunActivity:FindFormerNamesActivity   | 0.009              | 1     |  | a2a1ed5c9fa429d5d02d008bfd50986a |
|  | rocky-mountain-ruby-demo | RunActivity:FindJurisdictionsActivity | 0.023              | 1     |  | 8cfcb0c5467e66a5ecb01353cd9ddfbb |

UP AND COMERS

# DBOS

```
(.venv) → python-worker git:(main) x python3 start_background_check.py "Harry Kane"
Enqueued workflow 49e557a7-544f-4a54-906f-2d01607f81f4 for 'Harry Kane', waiting for it to complete...

former_names: Found former names for Harry Kane: Jane A. Smith, Jane Doe
jurisdictions: Jurisdictions to search for Harry Kane: Los Angeles County CA, Kings County NY
search_results: ['Search complete for Jane A. Smith in Los Angeles County, CA: no records found (via Python/DBOS worker)', 'Search complete for Jane Doe in Los Angeles County, CA: no records found (via Python/DBOS worker)', 'Search complete for Jane Doe in Kings County, NY: no records found (via Python/DBOS worker)', 'Search complete for Harry Kane in Los Angeles County, CA: no records found (via Python/DBOS worker)']
evaluation: Evaluation: all 4 searches clear, no adverse records found
comparison: Comparison: evaluation matches customer's stated assessment criteria (Evaluation: all 4 searches clear, no adverse records found)
```



# Election theme

| Candidate     | Async Workflow Job Library                               |
|---------------|----------------------------------------------------------|
| Incumbent     | Sidekiq                                                  |
| Challenger    | Temporal                                                 |
| Up and comers | Postgres or SQLite-backed<br>Durable Execution Libraries |

